2. FSR Pipeline Production Deployment & Backfill Plan

Pipeline Overview

The FSR pipeline ingests Field Service Report PDFs from Unity Catalog Volumes, extracts metadata via LLM, generates vector embeddings, and syncs to a Vector Search index for RAG retrieval.

Repo: GEV-SoX-DataBricks/pw_sdg_ai_ser_repo

Step	Notebook	What it does
DDL	`silver/src/ddl/nb_sdg_fsr_ddl`	Creates/ensures all Delta tables exist (idempotent)
P1 - Metadata	`silver/src/etl/nb_sdg_fsr_metadata`	Scans volumes for new PDFs, extracts metadata, normalizes via LLM, enriches from IBAT/EventVision/PSOT
P2 - Chunks	`gold/src/etl/nb_sdg_fsr_chunks`	Claims pending docs, extracts full text, chunks, generates embeddings, writes to Delta, syncs to Vector Search
Validate	`silver/src/validation/nb_sdg_fsr_validate`	41+ automated checks on data quality and pipeline consistency

Job Design

Five Databricks jobs cover setup, incremental runs, backfill, and ad-hoc validation:

Job Name (from YAML)	YAML File	Tasks	Schedule	Use case
`PW_SDG_FSR_DDL_Provision`	pw_sdg_fsr_ddl.yml	ddl	Manual	First-time setup, schema changes, or controlled reset
`PW_SDG_FSR_Ingestion`	`pw_sdg_fsr_ingestion.yml`	P1 P2 Validate	Scheduled (daily/weekly)	Steady-state: processes new PDFs as they arrive
`PW_SDG_FSR_Metadata_Backfill`	pw_sdg_fsr_metadata.yml	P1	Manual	Bulk metadata extraction for large initial loads
`PW_SDG_FSR_Chunking_Backfill`	pw_sdg_fsr_chunks.yml	P2	Manual	Bulk chunking/embedding — runs in parallel with metadata backfill
`PW_SDG_FSR_Validation`	pw_sdg_fsr_validate.yml	Validate	Manual	Ad-hoc health checks after backfill or fixes

Job Topology

Job	Tasks	Schedule	Purpose
PW_SDG_FSR_DDL_Provision	ddl	Manual	Infra-only — creates/ensures all Delta tables exist (idempotent)
PW_SDG_FSR_Ingestion	P1 P2 validate	Scheduled (daily/weekly)	Incremental — processes new PDFs as they land in the volume
PW_SDG_FSR_Metadata_Backfill	P1	Manual	Bulk metadata extraction for large initial loads
PW_SDG_FSR_Chunking_Backfill	P2	Manual	Bulk chunking/embedding — run in parallel with metadata backfill
PW_SDG_FSR_Validation	validate	Manual	Ad-hoc health checks — run after backfill batches, fixes, or on demand

Workflow files are now split into per-job YAMLs under `silver/src/workflows/fsr/`.

Topology diagram:

Incremental: P1 P2 Validate (single job, sequential)

Backfill: P1 (hours) (Job 3 — Metadata Backfill)
P2 (concurrent) (Job 4 — Chunking Backfill, runs alongside)
Validate (Job 5 — ad-hoc)

DDL: ddl (Job 1 — run once for setup/schema changes)

Tables

Table	Table Name	DEV	PROD
Metadata Registry	`biz_metadata_field_service_report`	`vaid.ai_sot_field_service_report`	`vaip.ai_sot_field_service_report`
Chunks	`vec_field_service_report`	`vaid.ai_std_con_field_service_report`	`vaip.ai_std_con_field_service_report`
Run Log	`fsr_run_log`	`vaid.ai_sot_field_service_report`	`vaip.ai_sot_field_service_report`
DQ Log	`fsr_data_quality_log`	`vaid.ai_sot_field_service_report`	`vaip.ai_sot_field_service_report`
Vector Search Index	`vs_vec_field_service_report`	`vaid.ai_std_con_field_service_report`	`vaip.ai_std_con_field_service_report`
VS Endpoint	`pw-ser-sdg-vector-search`	DEV workspace	PROD workspace

Input Sources

Source	Type	DEV	PROD
FieldVision FSRs	PDF Volume	`/Volumes/viud/ing_ud_fieldvision/fv_field_service_report`	`/Volumes/viup/ing_ud_fieldvision/fv_field_service_report`
Manual FSRs (eCRT)	PDF Volume	`/Volumes/viud/ing_ud_fsr_manual/manual_field_service_report/ecrt_reports`	`/Volumes/viup/ing_ud_fsr_manual/manual_field_service_report/ecrt_reports`
IBAT Equipment	Reference Table	`vgpp.prm_std_views.ibat_equipment_mst`	`vgpp.prm_std_views.ibat_equipment_mst`
Event Vision	Reference Table	`vgpp.fsr_std_views.eventmgmt_event_vision_sot`	`vgpp.fsr_std_views.eventmgmt_event_vision_sot`
FSR PDF Reference	Reference Table	`vgpp.fsr_std_views.fsr_pdf_ref`	`vgpp.fsr_std_views.fsr_pdf_ref`
PSOT (Outage/Tech)	Reference Table	`vgpp.fsr_std_views.fsr_field_vision_field_services_report_psot`	`vgpp.fsr_std_views.fsr_field_vision_field_services_report_psot`

Job Topology

Job	Tasks	Schedule	Airflow Job	Purpose
`FSR_Ingestion`	`ddl P1 P2 validate`	Scheduled (daily /weekly)		Incremental — processes new PDFs as they land in the volume
`FSR_Metadata_Backfill`	`ddl P1`	Manual		Bulk metadata extraction for large initial loads
`FSR_Chunking_Backfill`	`P2`	Manual		Bulk chunking/embedding — run in parallel with metadata backfill
`FSR_Validation`	`validate`	Manual		Ad-hoc health checks — run after backfill batches, fixes, or on demand

Airflow Jobs

Airflow Environment

Prod:

QA:

Dev:

Airflow env: bu-dna-dao-mwaa-gpr-02

airflow dag id -> databricks workflow

ser.pw_sdg_fsr_ingestion -> PW_SDG_FSR_Ingestion (scheduled 7 AM EST)

ser.pw_sdg_fsr_metadata_backfill->PW_SDG_FSR_Metadata_Backfill

ser.pw_sdg_fsr_chunking_backfill->PW_SDG_FSR_Chunking_Backfill

ser.pw_sdg_fsr_ddl_provision ->PW_SDG_FSR_DDL_Provision

ser.pw_sdg_fsr_validation ->PW_SDG_FSR_Validation